

Independent Smooth Signed Radial Kernel

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Approach

The new method learns a radial kernel from scratch instead of initializing from a previously trained kernel. but the profile starts from a generic decaying curve. It is parameterized so that f_r stays nonnegative, nonincreasing, and unit-norm during training. The normalization used here is $\|f\|_2 = 1$ during optimization. This prevents unstable growth and makes the profile easier to interpret.

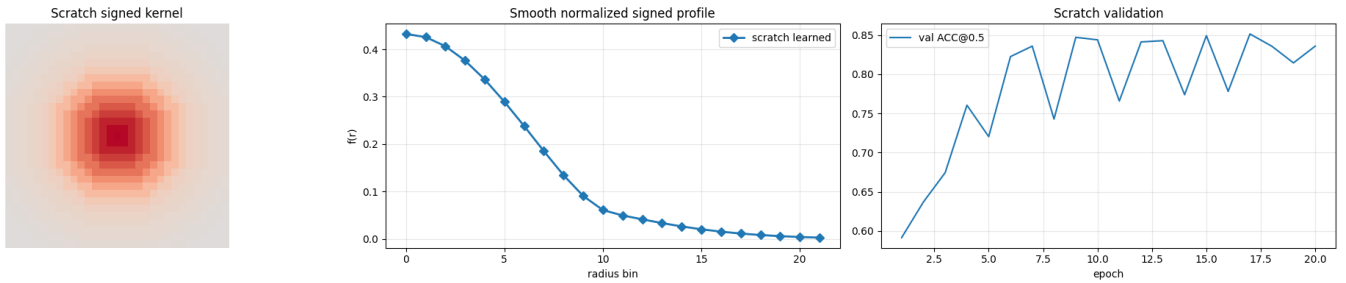
The response is also changed from **abs_max** to a signed top- k mean. The model computes the convolution map, keeps the strongest locations by absolute value, and averages their signed responses. I made this change to make the classification depend more on the learned kernel shape and less on a single magnitude peak.

The loss adds only the constraints needed to keep the profile well-behaved:

- smoothness between neighboring shells,
- curvature regularization to suppress spikes,
- radial decay to reduce large outer-shell values,
- a final-shell penalty to push the tail toward zero.

Test accuracy = 0.848.

Test AUC = 0.918725.



Independent smooth normalized signed radial kernel inference | stride=8 Test AUC=0.919 Test ACC@0.5=0.848

